

Database Project Report

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snickr: A Web-Based Collaboration System

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Introduction

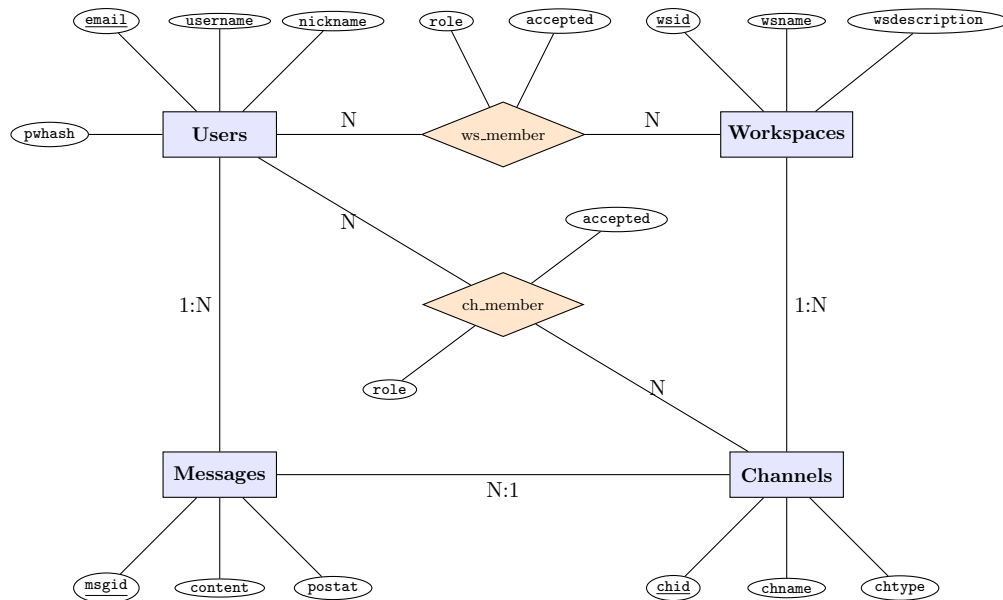
This report documents the design and implementation of *snickr*, a web-based collaboration system similar to Slack. The system allows users to create workspaces, organize communication through public, private, and direct channels, and exchange messages within those channels.

The database is built on PostgreSQL and consists of six relations: **users**, **workspaces**, **workspace_members**, **channels**, **channel_members**, and **messages**. Access control is enforced at the application level rather than through database permissions or views — a single DBMS account is used, and the application restricts content visibility based on workspace and channel membership.

This report covers the schema design with ER diagrams and relational translation (Section a), schema creation SQL (Section b), seven required queries with test results (Section c), and a discussion of the test data used to validate the queries (Section d).

(a) Database Schema Design

(a.1) ER Diagram



(a.2) Relational Schema

The ER diagram translates into the following six relations:

1. **users**(email, username, nickname, pwhash, createdat, updatedat)
2. **workspaces**(wsid, wsname, wsdescription, createdat, updatedat)
3. **workspace_members**(wsid, email, role, accepted, createdat, updatedat)
 - $\text{wsid} \rightarrow \text{workspaces}(\text{wsid})$
 - $\text{email} \rightarrow \text{users}(\text{email})$
 - $\text{role} \in \{\text{creator}, \text{admin}, \text{member}\}$
4. **channels**(chid, wsid, chname, chtype, createdat, updatedat)
 - $\text{wsid} \rightarrow \text{workspaces}(\text{wsid})$
 - $\text{chtype} \in \{\text{public}, \text{private}, \text{direct}\}$
5. **channel_members**(chid, email, role, accepted, createdat, updatedat)
 - $\text{chid} \rightarrow \text{channels}(\text{chid})$
 - $\text{email} \rightarrow \text{users}(\text{email})$
 - $\text{role} \in \{\text{creator}, \text{member}\}$
6. **messages**(msgid, chid, email, content, postat)
 - $\text{chid} \rightarrow \text{channels}(\text{chid})$
 - $\text{email} \rightarrow \text{users}(\text{email})$

(a.3) Design Assumptions

- **Creator is stored in membership tables, not in workspace/channel tables.** The creator is the first member inserted with role = 'creator'. This avoids redundancy and simplifies queries — no need to join an extra field when checking roles.
- **Invitations are tracked via the accepted field.** When a user is invited, a record is inserted with `accepted = FALSE`. Upon acceptance, it is updated to `TRUE`. This eliminates the need for a separate invitations table.
- **Public channels still require membership records.** Users in a workspace *can* join public channels, but are not automatically members. They must explicitly join, at which point a record is inserted with `accepted = TRUE`.
- **Passwords are stored as hashes.** The `pwhash` field stores a hashed password rather than plaintext for security.
- **Timestamps are recorded for all actions.** All tables include `createdat` and `updatedat` fields (messages use `postat` instead).
- **Access control is enforced at the application level,** not through database permissions or views, as specified in the project requirements.

(a.4) Space Efficiency

- **No redundant creator columns.** Instead of storing a separate `creator_email` in the `workspaces` and `channels` tables, we encode the creator as a membership record with `role = 'creator'`. This avoids duplicating user references.
- **Composite primary keys** on `workspace_members` and `channel_members` eliminate the need for surrogate ID columns in junction tables.
- **VARCHAR with appropriate limits** is used instead of fixed-length `CHAR` fields, so shorter values do not waste space.
- **SERIAL auto-increment** is used for `wsid`, `chid`, and `msgid` (4-byte integers), which is more compact than UUIDs.
- **No separate invitations table.** The `accepted` boolean field on membership tables tracks invitation state in-place, avoiding an extra table and the joins it would require.

(b) Schema Creation

The following SQL creates the schema in PostgreSQL with all key, foreign key, and check constraints:

```
CREATE TABLE users (
    email        VARCHAR(255) PRIMARY KEY,
    username     VARCHAR(50)  NOT NULL,
    nickname     VARCHAR(50),
    pwhash       VARCHAR(255) NOT NULL,
    createdat    TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updatedat    TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE workspaces (
    wsid         SERIAL        PRIMARY KEY,
    wsname       VARCHAR(100)  NOT NULL,
    wsdescription TEXT,
    createdat    TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updatedat    TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE workspace_members (
    wsid         INT           NOT NULL REFERENCES workspaces(wsid),
    email        VARCHAR(255) NOT NULL REFERENCES users(email),
    role         VARCHAR(10)   NOT NULL CHECK (role IN ('creator', 'admin', 'member')),
    accepted     BOOLEAN       NOT NULL DEFAULT FALSE,
    createdat    TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updatedat    TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (wsid, email)
);

CREATE TABLE channels (
    chid         SERIAL        PRIMARY KEY,
    wsid         INT           NOT NULL REFERENCES workspaces(wsid),
    chname       VARCHAR(100),
    chtype       VARCHAR(10)   NOT NULL CHECK (chtype IN ('public', 'private', 'direct')),
    createdat    TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updatedat    TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE channel_members (
    chid         INT           NOT NULL REFERENCES channels(chid),
    email        VARCHAR(255) NOT NULL REFERENCES users(email),
    role         VARCHAR(10)   NOT NULL CHECK (role IN ('creator', 'member')),
    accepted     BOOLEAN       NOT NULL DEFAULT FALSE,
    createdat    TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updatedat    TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (chid, email)
);

CREATE TABLE messages (
    msgid        SERIAL        PRIMARY KEY,
    chid         INT           NOT NULL REFERENCES channels(chid),
    email        VARCHAR(255) NOT NULL REFERENCES users(email),
    content      TEXT          NOT NULL,
    postat       TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP
);
```

(c) Queries

(1) Create a new user account

```
CREATE OR REPLACE PROCEDURE create_user(  
    p_email    VARCHAR,  
    p_username VARCHAR,  
    p_nickname VARCHAR,  
    p_pwhash   VARCHAR  
)  
LANGUAGE SQL  
AS $$  
    INSERT INTO users (email, username, nickname, pwhash)  
    VALUES (p_email, p_username, p_nickname, p_pwhash);  
$$;  
  
-- Example call:  
CALL create_user('gz1234@nyu.edu', 'gracezhu', 'Grace', 'a1b2c3d4e5f6');
```

Test: Calling with a new email succeeds and inserts a row into `users`. Calling with a duplicate email raises a primary key violation, confirming the uniqueness constraint.

(2) Create a new public channel (with authorization check)

```
CREATE OR REPLACE PROCEDURE create_public_channel(  
    p_wsids INT,  
    p_email  VARCHAR,  
    p_chname VARCHAR  
)  
LANGUAGE plpgsql  
AS $$  
DECLARE  
    v_chid INT;  
BEGIN  
    -- Check if the user is an accepted member of the workspace  
    IF NOT EXISTS (  
        SELECT 1 FROM workspace_members  
        WHERE wsid = p_wsids AND email = p_email AND accepted = TRUE  
    ) THEN  
        RAISE EXCEPTION 'User % is not an accepted member of workspace %',  
            p_email, p_wsids;  
    END IF;  
  
    -- Create the public channel  
    INSERT INTO channels (wsid, chname, chtype)  
    VALUES (p_wsids, p_chname, 'public')  
    RETURNING chid INTO v_chid;  
  
    -- Add the creator as a channel member  
    INSERT INTO channel_members (chid, email, role, accepted)  
    VALUES (v_chid, p_email, 'creator', TRUE);  
END;  
$$;  
  
-- Example call:  
CALL create_public_channel(1, 'az1234@nyu.edu', 'announcements');
```

Test: Calling with an accepted workspace member succeeds, creating a new channel and adding the user as channel creator. Calling with a non-member (e.g., Eve, who has a pending invite) raises the exception `User % is not an accepted member of workspace %`.

(3) For each workspace, list all current administrators

```
SELECT w.wsid, w.wsname, wm.email, u.username, wm.role
FROM workspaces w
JOIN workspace_members wm ON w.wsid = wm.wsid
JOIN users u ON wm.email = u.email
WHERE wm.role IN ('creator', 'admin') AND wm.accepted = TRUE
ORDER BY w.wsid, wm.role;
```

Result:

wsid	wsname	email	username	role
1	CS Department	bs5678@nyu.edu	bobsmith	admin
1	CS Department	az1234@nyu.edu	alicezhang	creator
2	Tandon Robotics Club	cw9012@nyu.edu	carolwang	admin
2	Tandon Robotics Club	bs5678@nyu.edu	bobsmith	creator

(4) Pending invitations per public channel (older than 5 days)

```
SELECT c.chid, c.chname, COUNT(cm.email) AS pending_count
FROM channels c
LEFT JOIN channel_members cm ON c.chid = cm.chid
AND cm.accepted = FALSE
AND cm.createdat < CURRENT_TIMESTAMP - INTERVAL '5 days'
WHERE c.wsid = 1 AND c.chtype = 'public'
GROUP BY c.chid, c.chname
ORDER BY c.chid;
```

Result:

chid	chname	pending_count
1	general	1

(5) List all messages in a channel in chronological order

```
CREATE OR REPLACE FUNCTION list_channel_messages(p_chid INT)
RETURNS TABLE (
    msgid INT, email VARCHAR, username VARCHAR,
    content TEXT, postat TIMESTAMP
)
LANGUAGE SQL
AS $$
    SELECT m.msgid, m.email, u.username, m.content, m.postat
    FROM messages m
    JOIN users u ON m.email = u.email
    WHERE m.chid = p_chid
    ORDER BY m.postat ASC;
$$;

-- Example call:
SELECT * FROM list_channel_messages(1);
```

Result:

msgid	email	username	content	postat
1	az1234@nyu.edu	alicezhang	Welcome to the CS Department workspace!	2026-04-01 09:00
2	bs5678@nyu.edu	bobsmith	Thanks Alice! Excited to be here.	2026-04-01 09:05
3	cw9012@nyu.edu	carolwang	Is the new curriculum perpendicular to the old one?	2026-04-01 09:10
4	az1234@nyu.edu	alicezhang	Not exactly perpendicular, but quite different.	2026-04-01 09:15

(6) List all messages posted by a particular user

```
SELECT m.msgid, c.chname, m.content, m.postat
FROM messages m
JOIN channels c ON m.chid = c.chid
WHERE m.email = 'az1234@nyu.edu'
ORDER BY m.postat DESC;
```

Result:

msgid	chname	content	postat
7	(DM)	Bob, can you handle the seminar logistics?	2026-04-03 14:00
5	hiring-committee	We need to review the new candidates.	2026-04-02 10:00
4	general	Not exactly perpendicular, but quite different.	2026-04-01 09:15
1	general	Welcome to the CS Department workspace!	2026-04-01 09:00

(7) Search accessible messages by keyword

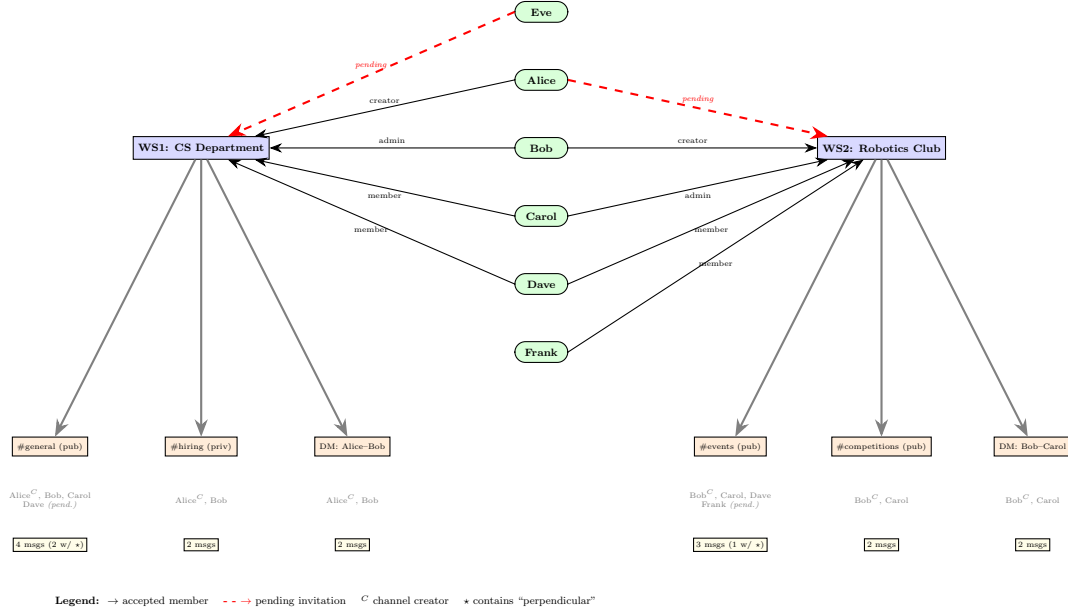
```
CREATE OR REPLACE FUNCTION search_accessible_messages(
    p_email VARCHAR, p_keyword VARCHAR
)
RETURNS TABLE (
    msgid INT, chname VARCHAR, author VARCHAR,
    content TEXT, postat TIMESTAMP
) LANGUAGE SQL AS $$
    SELECT m.msgid, c.chname, m.email AS author,
           m.content, m.postat
    FROM messages m
    JOIN channels c ON m.chid = c.chid
    JOIN channel_members cm ON c.chid = cm.chid
    JOIN workspace_members wm
        ON c.wsid = wm.wsid AND wm.email = p_email
    WHERE cm.email = p_email
           AND cm.accepted = TRUE
           AND wm.accepted = TRUE
           AND m.content ILIKE '%' || p_keyword || '%'
    ORDER BY m.postat DESC;
$$;
-- Example call:
SELECT * FROM search_accessible_messages('az1234@nyu.edu', 'perpendicular');
```

Result:

msgid	chname	author	content	postat
4	general	az1234@nyu.edu	Not exactly perpendicular, but quite different.	2026-04-01 09:15
3	general	cw9012@nyu.edu	Is the new curriculum perpendicular to the old one?	2026-04-01 09:10

(d) Test Data

The diagram below illustrates the structure of our test data. Solid arrows represent accepted memberships; dashed arrows represent pending invitations. Messages containing the keyword “perpendicular” are marked with ★.



Data Structure Discussion

- **6 users** span two workspaces with overlapping membership: Alice, Bob, Carol, and Dave appear in both, while Eve and Frank each belong to only one — ensuring cross-workspace queries return correct results.
- **2 workspaces** each contain 3 channels covering all three types (public, private, direct), so queries that filter by channel type are exercised.
- **Role diversity:** Alice is a workspace creator and channel creator; Bob is both a workspace creator (Robotics Club) and an admin (CS Department). This tests query (3) which lists administrators.
- **Pending invitations:** Eve has not accepted her workspace invite; Dave and Frank have not accepted their channel invites (created >5 days ago). This tests query (4).
- **15 messages** are distributed across all 6 channels. Three messages contain the keyword “perpendicular” in two different channels across both workspaces, testing query (7)’s keyword search with access control.